

Grade 11 Maths Unit 1: Relations and Functions

1.1 Relations

Competency

By the end of this sub-unit, students will be able to:

- **Sketch the graph of a relation.**

1.2 Inverse of Relations and their Graphs

Competency

By the end of this sub-unit, students will be able to:

- **calculate the inverse of a given relation; and**
- **sketch the graph of the inverse of a relation.**

1. 3 Types of Power function

1.3.1 Power Functions with Their Graphs

Competency

By the end of this sub-unit, students will be able to:

- **define power functions;**
- **describe the properties of power functions in relation to their exponents;**
- **determine the domains and ranges of power functions;**
- **sketch the graphs of power functions;**

1.3.2 Modulus (Absolute value) Function

Competency

By the end of this sub-unit, students will be able to:

- **define Modulus Function (Absolute value Function);**
- **determine the domain and the range of modulus function;**
- **sketch the graph of a Modulus function;**

1.3.3 Signum Function

Competency

By the end of this sub-unit, students will be able to:

- **define the Signum function;**
- **determine the domain and range of Signum function;**
- **sketch the graph of the Signum function;**

1.3.4 The Greatest Integer (step or floor) Function

Competency

By the end of this sub-unit, students will be able to:

- **define the "Greatest Integer Function";**
- **determine the domain and range of the Greatest Integer function;**
- **use the concept of greatest integer function for billing of shipment (postal); and**
- **sketch the graph of the Greatest Integer function.**

1.4 Composition of Functions

Competency

By the end of this sub-unit, students will be able to:

- define the composition of function;
- determine the composite function given the component functions;
- determine the domain and the range of a composite function of two given functions; and
- compare the domain of component functions and composite function.

1.5 Inverse Functions and Their Graphs

Competency

By the end of this sub-unit, students will be able to:

- recognize real valued functions;
- describe the condition for the existence of inverse function;
- calculate inverse function for an invertible function;
- determine whether two given functions are inverses of each other or not;
- determine the domain and range of the inverse of a given function;
- sketch the graph of the inverse function; and

1.6 Applications of Relations and Functions

Competency

By the end of this sub-unit, students will be able to:

- use relation and function in solving real life problem